CMPG324

COMPUTER NETWORK

INDIVIDUAL PROJECT MILESTONE 2

CLIENT IMPLEMENTATION REVIEW

NAME: KATLEGO TAWANA

STUDENT NUMBER: 29290473

Section 1: Introduction

This document shows and explains how the design of Lesika Data Analytics' network is implemented. It covers the following:

- VLAN configurations.
- CR15-dual ISP resilience.
- The assigned networking challenge – Network troubleshooting.
- Testing results/evidence.
- DHCP and DNS services.
- Router-on-a-stick inter-VLAN routing.
- Wireless segmentation.

(Screenshots are provided as proof that the network functions properly.

Section 2: Implementation overview

The network was developed, created, and configured using Cisco Packet Tracer. The network's main components and configurations are:

- CR15: The network backup is implemented by means of a primary static route through ISP1 and a floating static route through ISP2.
- Edge Router(R1): This connects the network to two ISP routers by means of WAN connections.
- Core Switch(SW1): Uses trunk links to connect two access switches (SW2 and SW3).
- VLANS: VLAN10 for Data Analysts, VLAN20 for Data Scientists, VLAN30 for Management+Sales, VLAN40 for servers, VLAN50 for Printers, VLAN60 for Guest Wi-Fi, and VLAN99 for Network Management.
- DHCP and DNS: The server SRV-DHCP-DNS at 192.168.60.98 provides these services.

- DHCP relay: Configured via IP helper addresses on VLAN sub-interfaces, forwarding DHCP requests to SRV-DHCP-DNS.
- Wireless: The Wi-Fi network is made possible by two access points, AP-staff for staff and AP-Guest
- NAT overload: Configured on the Edge router to allow internal devices within the network to access the internet.
- Router-on-stick: To allow communication between the different VLANs, the edge router(R1) uses an 802.1Q sub-interface.
- Guest isolation: ACL 100 is implemented on/ in VLAN60 so that guests' devices are restricted from accessing the internal networks.

Section 3: Testing Evidence

1. Test 1: DHCP from wired VLAN10 PC

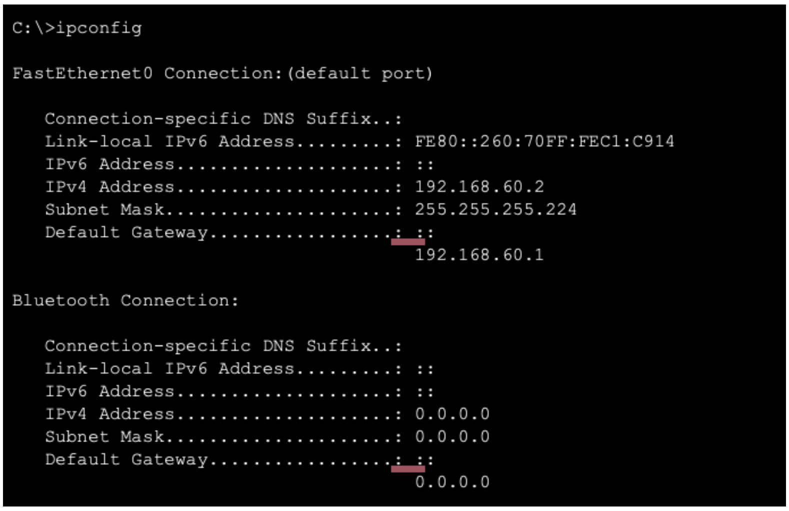
Device: PC-AN1

Command:ipconfig

Expectation: Ip address range from 192.168.60.2 -.30 , and with gateway 192.168.60.1

Outcome: PASS

Screenshot/proof:



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:70FF:FEC1:C914
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.60.2
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.60.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

2. Test 2: Ping default gateway from VLAN10

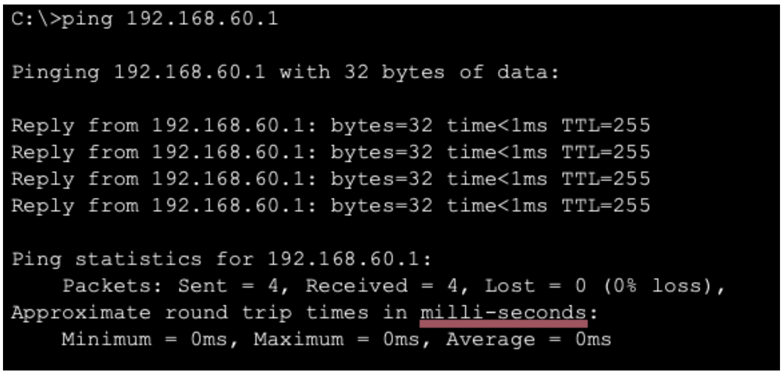
Device: PC-AN1

Command: ping 192.168.60.1

Expectation: 4/4 replies

Outcome: PASS

Screenshot/proof:



```
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. Test 3: Cross VLAN ping to servers

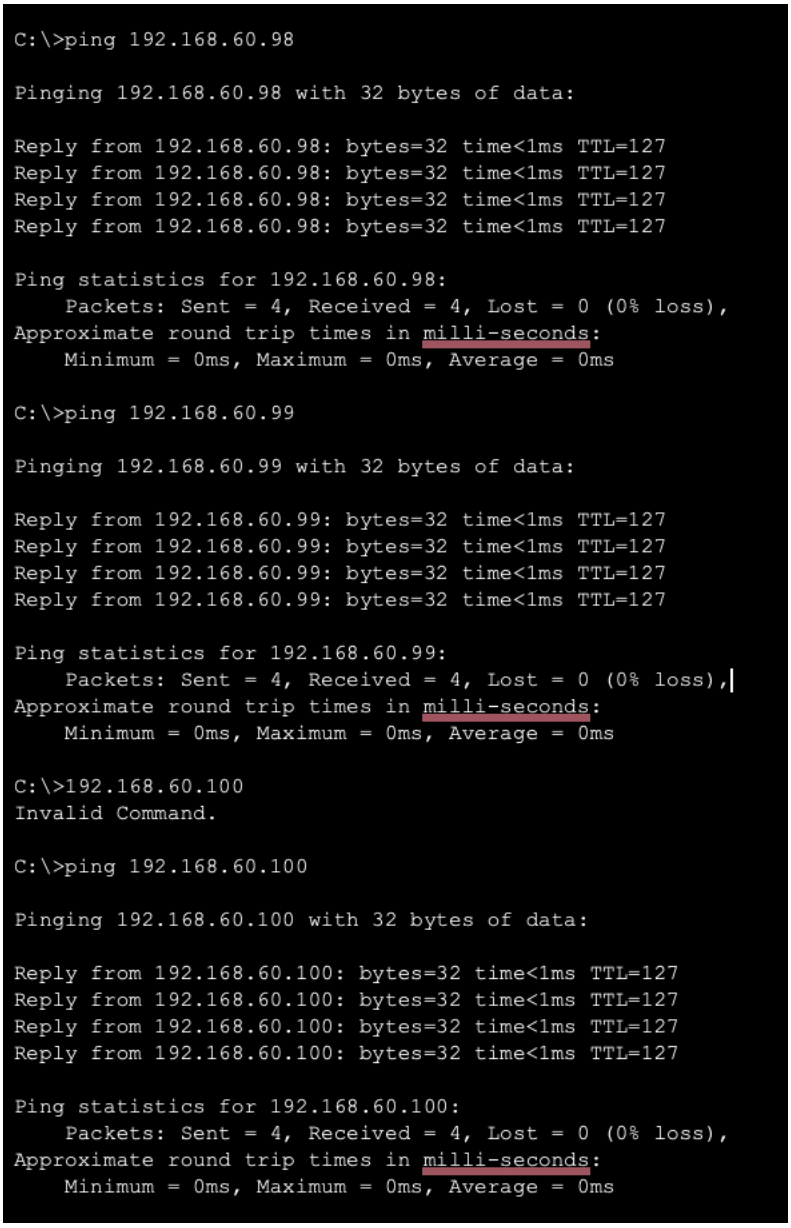
Device: PC-AN1

Command: ping 192.168.60.98, ping 192.168.60.99, ping 192.168.60.100

Expectation: all succeed

Outcome: PASS

Screenshot/proof:



```
C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.60.99

Pinging 192.168.60.99 with 32 bytes of data:

Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>192.168.60.100
Invalid Command.

C:\>ping 192.168.60.100

Pinging 192.168.60.100 with 32 bytes of data:

Reply from 192.168.60.100: bytes=32 time<1ms TTL=127
Reply from 192.168.60.100: bytes=32 time<1ms TTL=127
Reply from 192.168.60.100: bytes=32 time<1ms TTL=127
Reply from 192.168.60.100: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. Test 4: DNS Resolution

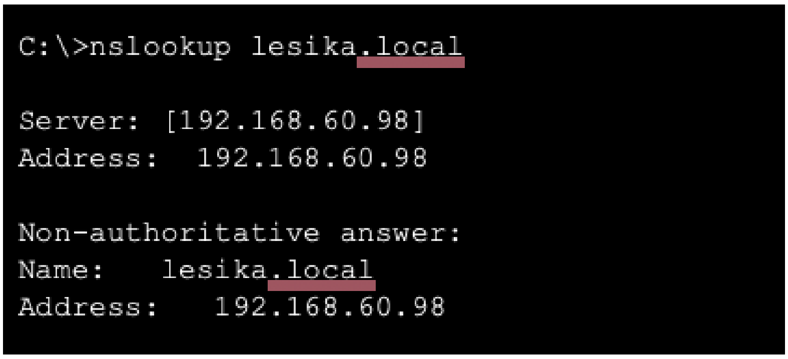
Device: PC-AN1

Command: nslookup lesika.local

Expectation: resolves to 192.168.60.98

Outcome: PASS

Screenshot/proof:



```
C:\>nslookup lesika.local

Server: [192.168.60.98]
Address:  192.168.60.98

Non-authoritative answer:
Name:   lesika.local
Address:   192.168.60.98
```

5. Test 5: Wireless connectivity for staff

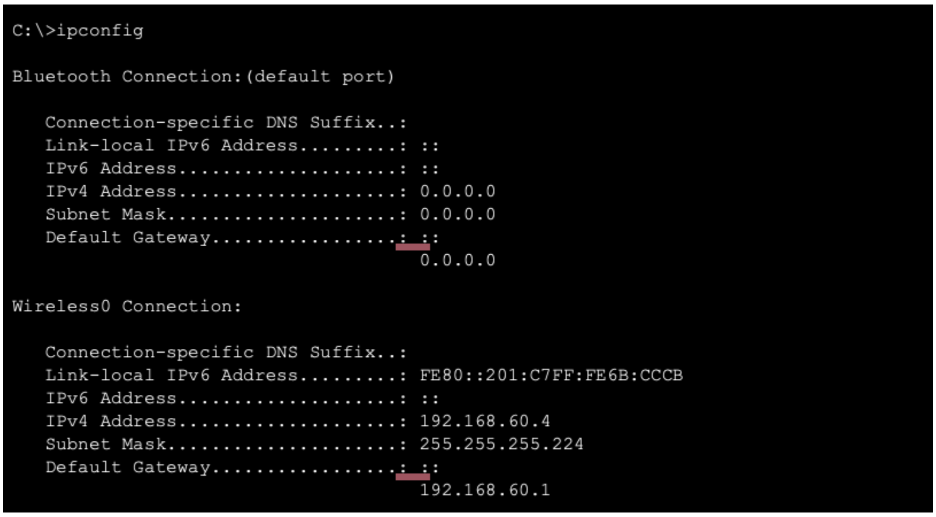
Device: LAPTOP-STAFF

Command: ipconfig(after connecting to SSID LDA-Staff)

Expectation: Ip in range between 192.168.60.2-.30, with gateway 192.168.60.1

Outcome: PASS

Screenshot/proof:



```
C:\>ipconfig

Bluetooth Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

Wireless0 Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:C7FF:FE6B:CCCB
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.60.4
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.60.1
```

6. Wireless connectivity for guests

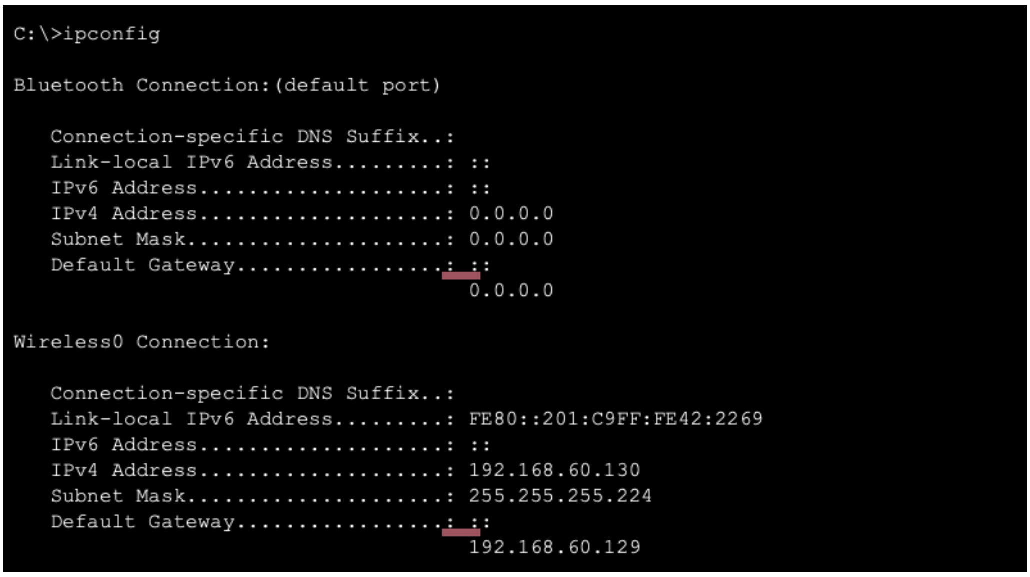
Device: LAPTOP-GUEST

Command: ipconfig(after connecting to SSID LDA-Guest)

Expectation: Ip in range between 192.168.60.130 -.158, with gateway 192.168.60.129

Outcome: PASS

Screenshot/proof:



```
C:\>ipconfig

Bluetooth Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

Wireless0 Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:C9FF:FE42:2269
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.60.130
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.60.129
```

7. Test 7: CR15- Backup route active

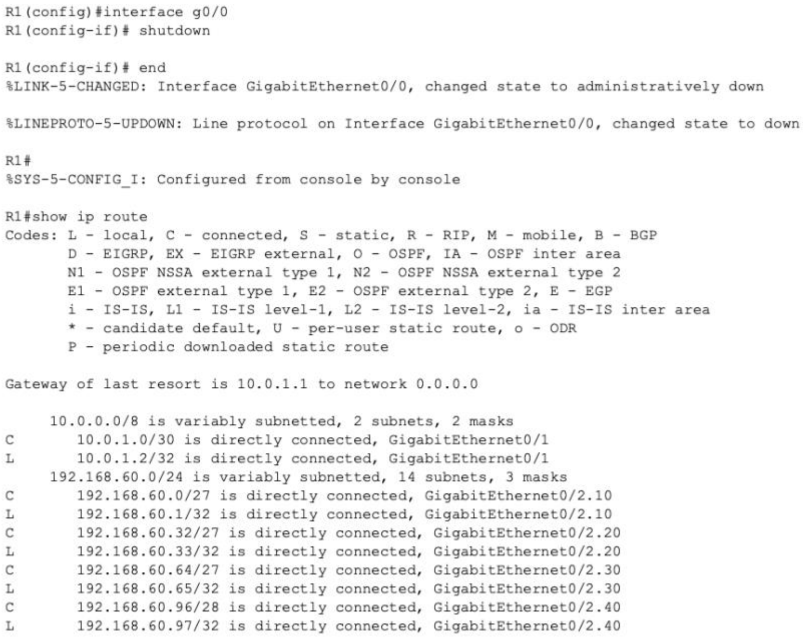
Device: R1

Command: show ip route after shutting down G0/0

Expectation: Gateway of last resort changes to ISP2(10.0.1.1)

Outcome: PASS

Screenshot/proof:



```
R1(config)#interface g0/0
R1(config-if)# shutdown

R1(config-if)# end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.1.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.2/32 is directly connected, GigabitEthernet0/1
     192.168.60.0/24 is variably subnetted, 14 subnets, 3 masks
C       192.168.60.0/27 is directly connected, GigabitEthernet0/2.10
L       192.168.60.1/32 is directly connected, GigabitEthernet0/2.10
C       192.168.60.32/27 is directly connected, GigabitEthernet0/2.20
L       192.168.60.33/32 is directly connected, GigabitEthernet0/2.20
C       192.168.60.64/27 is directly connected, GigabitEthernet0/2.30
L       192.168.60.65/32 is directly connected, GigabitEthernet0/2.30
C       192.168.60.96/28 is directly connected, GigabitEthernet0/2.40
L       192.168.60.97/32 is directly connected, GigabitEthernet0/2.40
```

8. Test 8: CR15-primary route restored

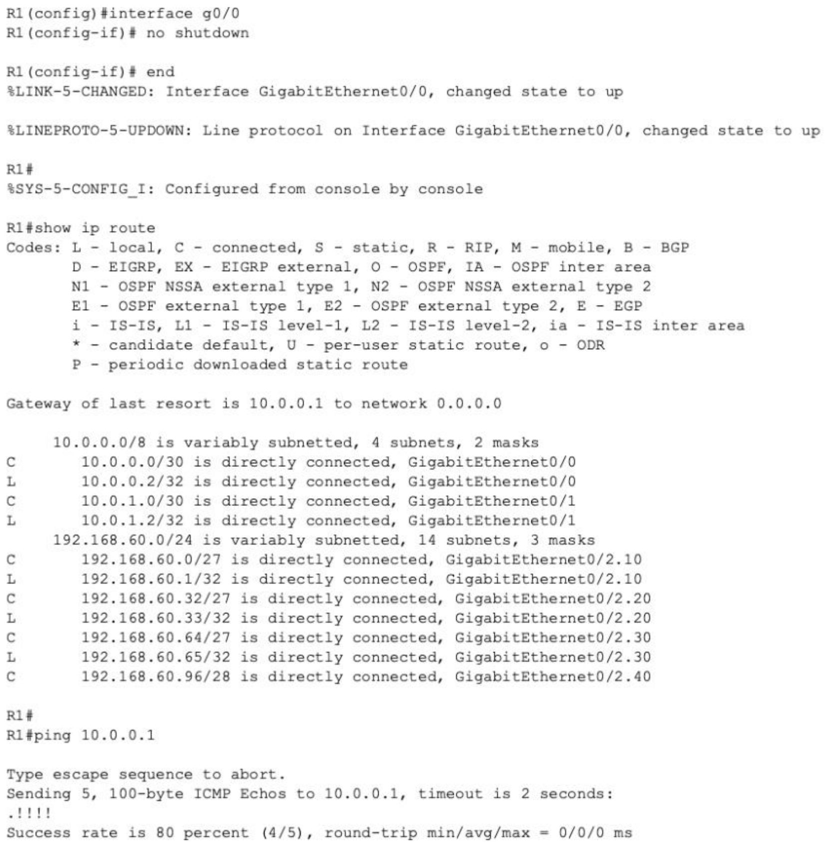
Device: R1

Command: show ip route after shutting down G0/0

Expectation: Gateway of last resort changes to ISP1(10.0.0.1)

Outcome: PASS

Screenshot/proof:



```
R1(config)#interface g0/0
R1(config-if)# no shutdown

R1(config-if)# end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.2/32 is directly connected, GigabitEthernet0/1
     192.168.60.0/24 is variably subnetted, 14 subnets, 3 masks
C       192.168.60.0/27 is directly connected, GigabitEthernet0/2.10
L       192.168.60.1/32 is directly connected, GigabitEthernet0/2.10
C       192.168.60.32/27 is directly connected, GigabitEthernet0/2.20
L       192.168.60.33/32 is directly connected, GigabitEthernet0/2.20
C       192.168.60.64/27 is directly connected, GigabitEthernet0/2.30
L       192.168.60.65/32 is directly connected, GigabitEthernet0/2.30
C       192.168.60.96/28 is directly connected, GigabitEthernet0/2.40

R1#
R1#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

9. Test 9: Guest Isolation

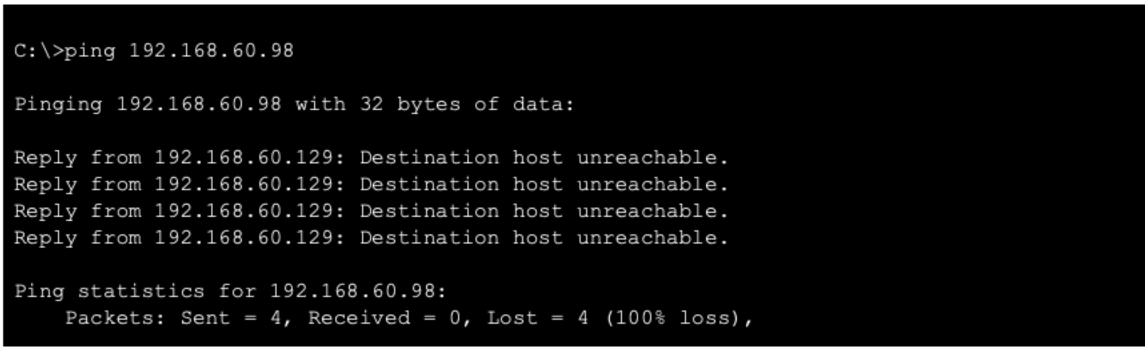
Device: LAPTOP-GUEST

Command: ping 192.168.60.98

Expectation: ping fails, ACL 100 will block guest to internal network traffic.

Outcome: PASS

Screenshot/proof:



```
C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Reply from 192.168.60.129: Destination host unreachable.
Reply from 192.168.60.129: Destination host unreachable.
Reply from 192.168.60.129: Destination host unreachable.
Reply from 192.168.60.129: Destination host unreachable.

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

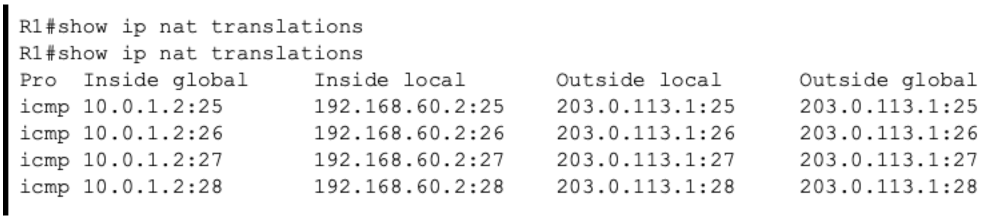
10. Test 10: NAT Translations

Device: R1

Command: show ip nat translations after PC-An1 pings an external address

Expectation: ICMP entries showing 192.168.60.2 translated to WAN interface address

Screenshot/proof:



```
R1#show ip nat translations
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.0.1.2:25       192.168.60.2:25    203.0.113.1:25     203.0.113.1:25
icmp 10.0.1.2:26       192.168.60.2:26    203.0.113.1:26     203.0.113.1:26
icmp 10.0.1.2:27       192.168.60.2:27    203.0.113.1:27     203.0.113.1:27
icmp 10.0.1.2:28       192.168.60.2:28    203.0.113.1:28     203.0.113.1:28
```

11. Test 11: VLAN Configuration on core switch

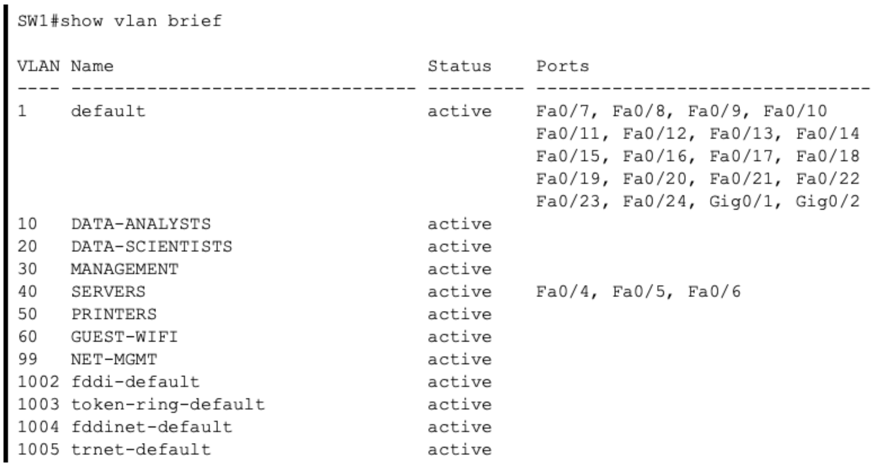
Device: SW1

Command: show vlan brief

Expectation: VLAN10,20,30,40,50,60,99 assigned to the correct ports

Outcome: PASS

Screenshot/proof:



```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   DATA-ANALYSTS                    active
20   DATA-SCIENTISTS                  active
30   MANAGEMENT                       active
40   SERVERS                          active    Fa0/4, Fa0/5, Fa0/6
50   PRINTERS                         active
60   GUEST-WIFI                       active
99   NET-MGMT                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

12. Test 12: Trunk configuration on core switch

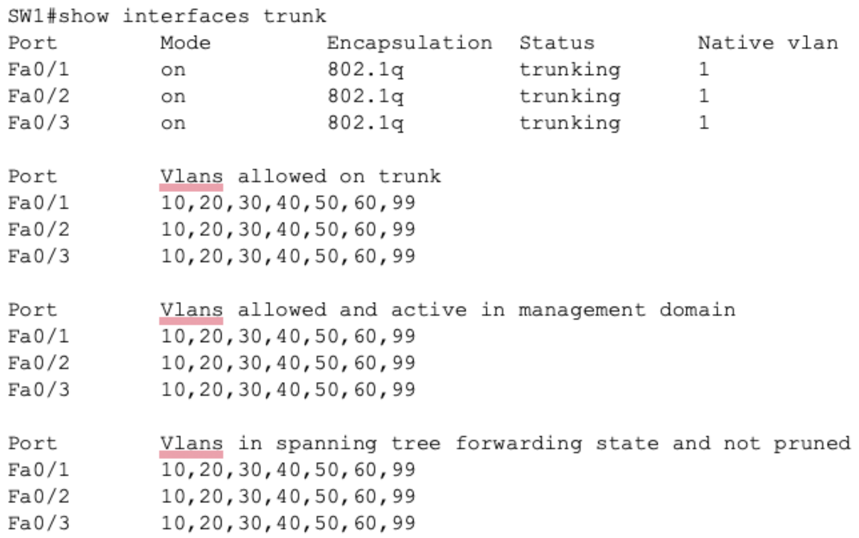
Device: SW1

Command: show interfaces trunk

Expectation: Fa0/1, Fa0/2, Fa0/3 in trunking mode

Outcome: PASS

Screenshot/proof:



```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99
```

13. Test 13: Router sub-interfaces on edge router

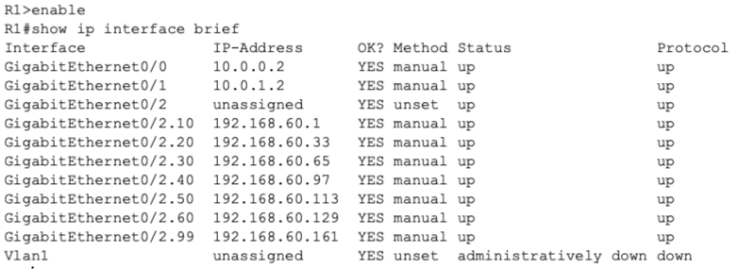
Device:R1

Command: show ip interface brief

Expectation: G0/2.10 to G0/2.99 protocol show up and are assigned correct IP adreesses

Outcome: PASS

Screenshot/proof:



```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.2        YES manual up                    up
GigabitEthernet0/1     10.0.1.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.10  192.168.60.1    YES manual up                    up
GigabitEthernet0/2.20  192.168.60.33   YES manual up                    up
GigabitEthernet0/2.30  192.168.60.65   YES manual up                    up
GigabitEthernet0/2.40  192.168.60.97   YES manual up                    up
GigabitEthernet0/2.50  192.168.60.113  YES manual up                    up
GigabitEthernet0/2.60  192.168.60.129  YES manual up                    up
GigabitEthernet0/2.99  192.168.60.161  YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

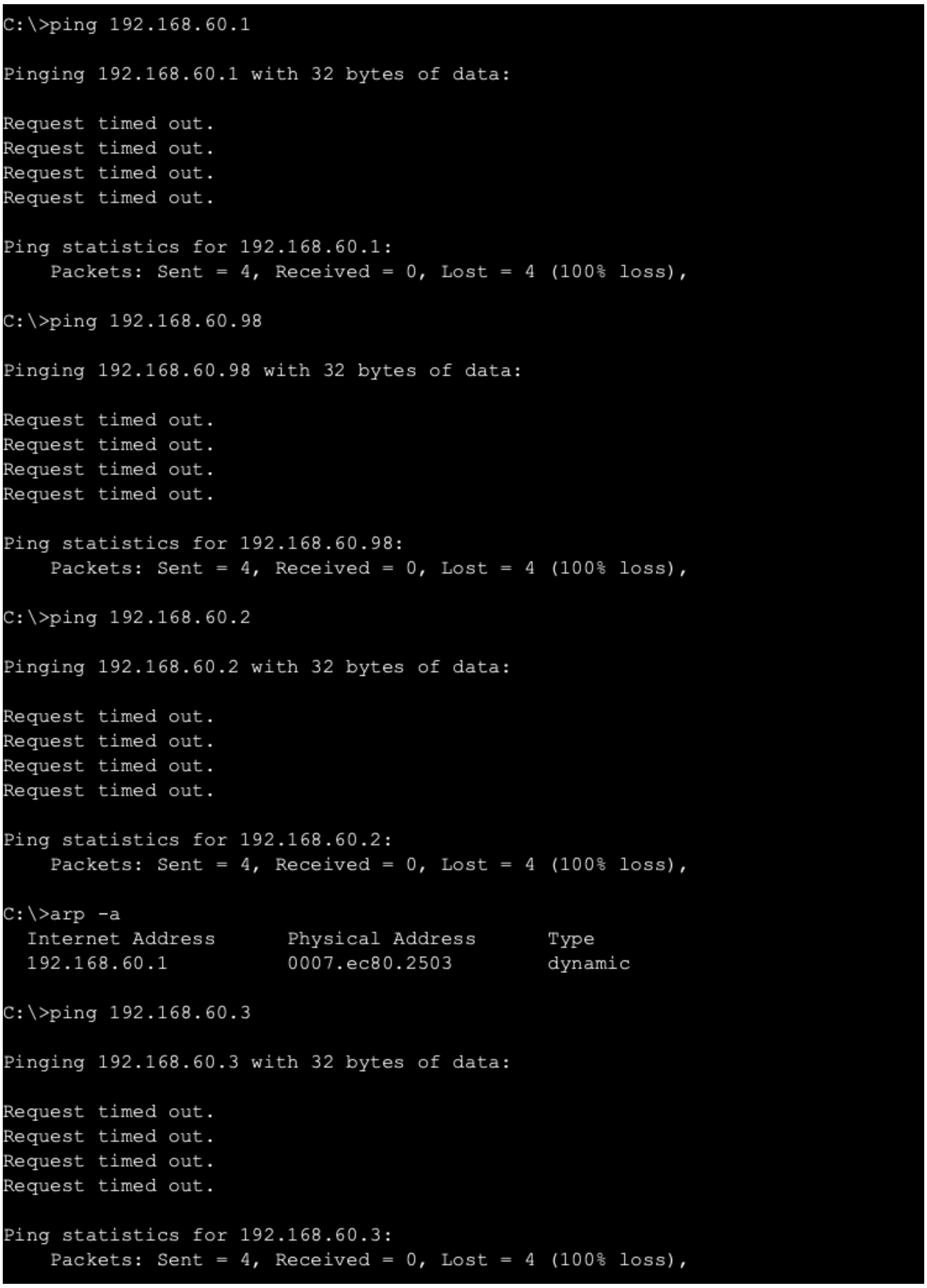
Section 4: Assigned Networking Challenge

Network Troubleshooting-fault isolation is the assigned networking challenge. The following 5 Realistic faults were introduced, diagnosed, and fixed:

1. Disabled trunk port between SW1 and SW2
2. Missing VLAN 20 from trunk permitted list.
3. Wrong IP address on a PC.
4. Missing IP helper address on VLAN 20 sub-interface.
5. Missing DNS A-record on the SRV-DHCP-DNS server.

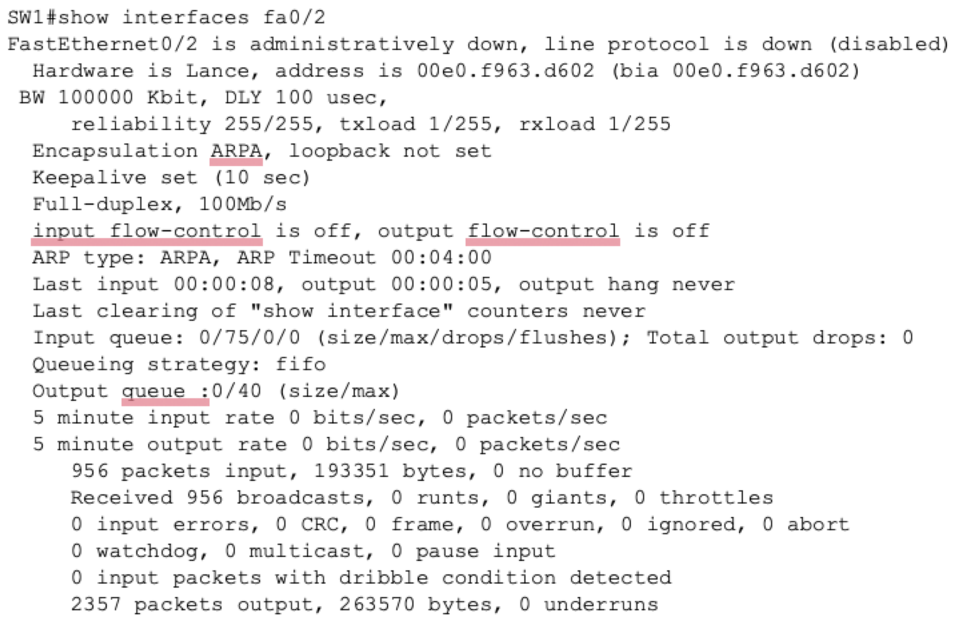
1. Fault 1:Disabled trunk port between SW1 and SW2

- Symptom: PC-AN1 can’t reach the gateway



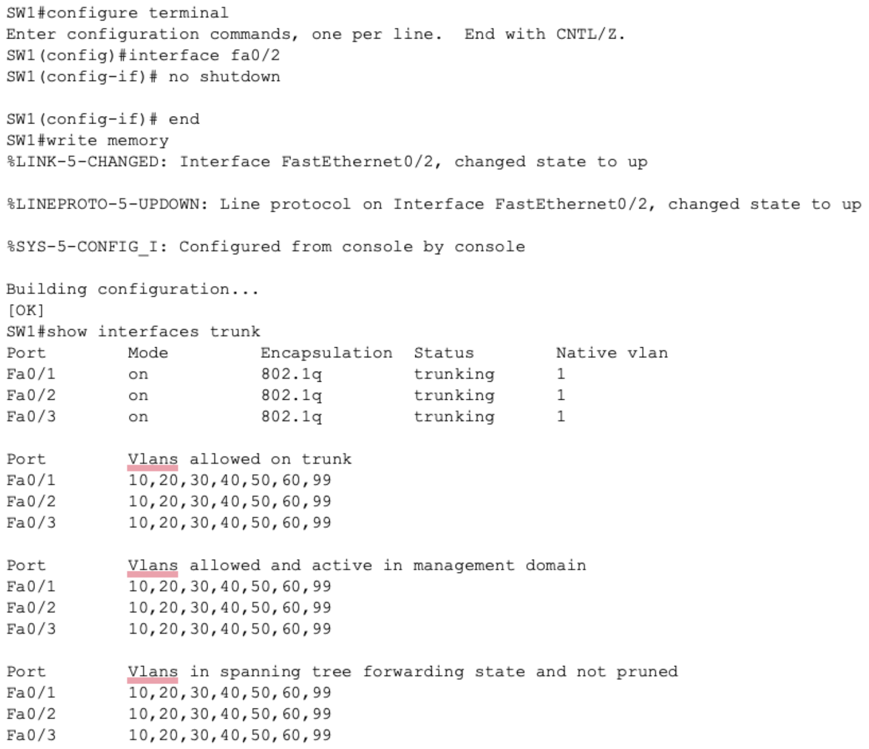
```
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.60.2

Pinging 192.168.60.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.60.1          0007.ec80.2503        dynamic

C:\>ping 192.168.60.3

Pinging 192.168.60.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Diagnostic: “show interfaces fa0/2 on the core switch(SW1) is administratively down.



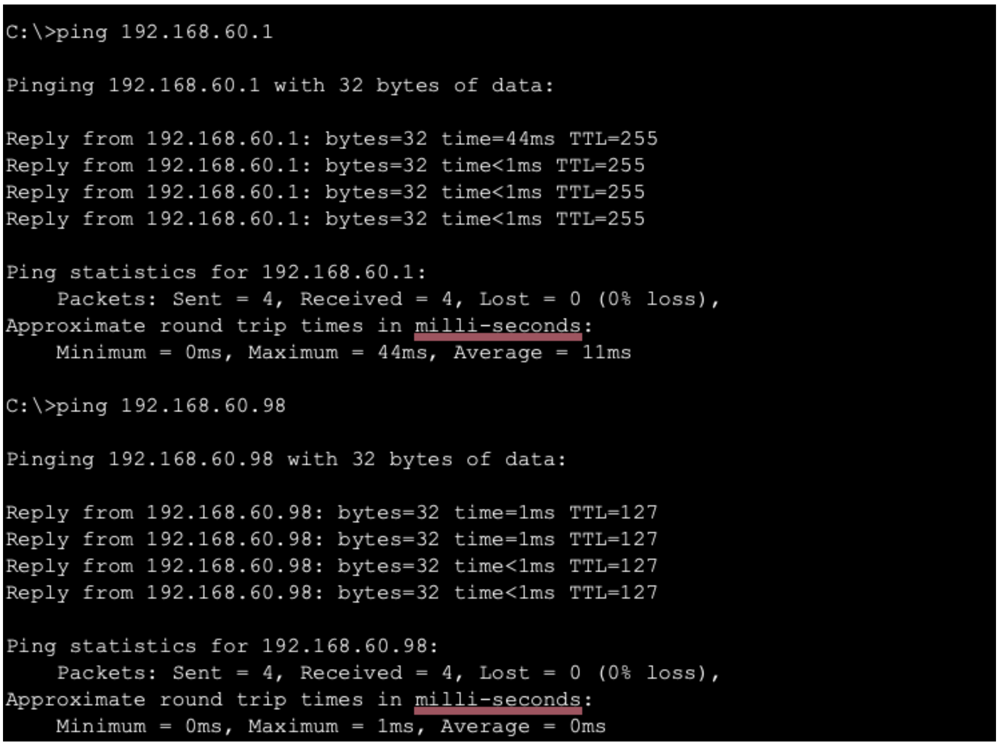
```
SW1#show interfaces fa0/2
FastEthernet0/2 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 00e0.f963.d602 (bia 00e0.f963.d602)
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
     2357 packets output, 263570 bytes, 0 underruns
```

- Fix: “no shutdown’ on core switch Fa0/2



```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)# no shutdown

SW1(config-if)# end
SW1#write memory
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99
```

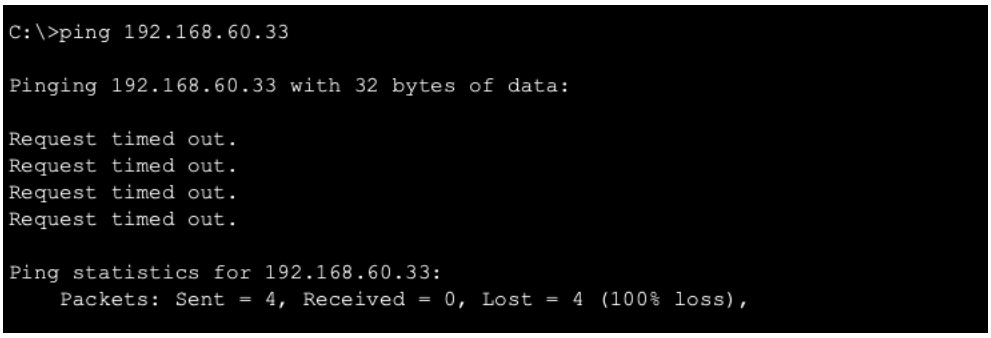
- Verification: Pings succeed after the fix:



```
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Reply from 192.168.60.1: bytes=32 time=44ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 44ms, Average = 11ms

C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Reply from 192.168.60.98: bytes=32 time=1ms TTL=127
Reply from 192.168.60.98: bytes=32 time=1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

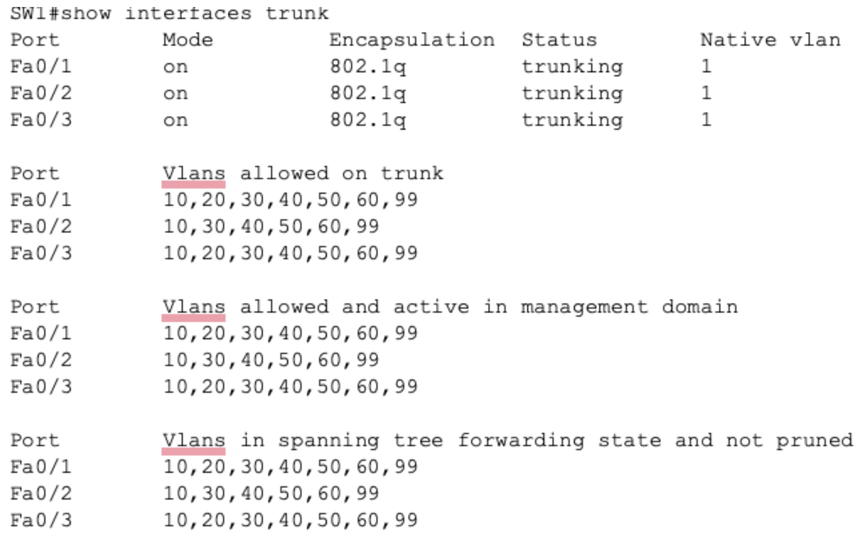
2. Fault 2: VLAN 20 missing from trunk permitted list.

- Symptom: Data Scientists can’t reach their gateway or any VLAN = PC-DS1 can’t reach the gateway (192.168.60.33) and the file server (192.168.60.98).



```
C:\>ping 192.168.60.33

Pinging 192.168.60.33 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.33:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

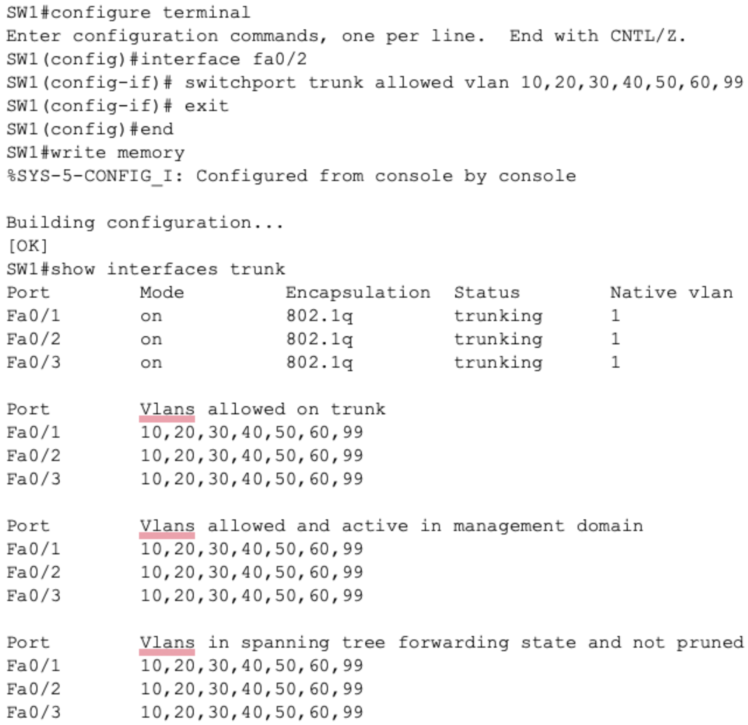
- Diagnostic: IP configuration shows address on an incorrect subnet = “show interfaces trunk on core switch shows fa0/2 allowing VLANs as 10,30,40,50,60,99., VLAN is missing.



```
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99
```

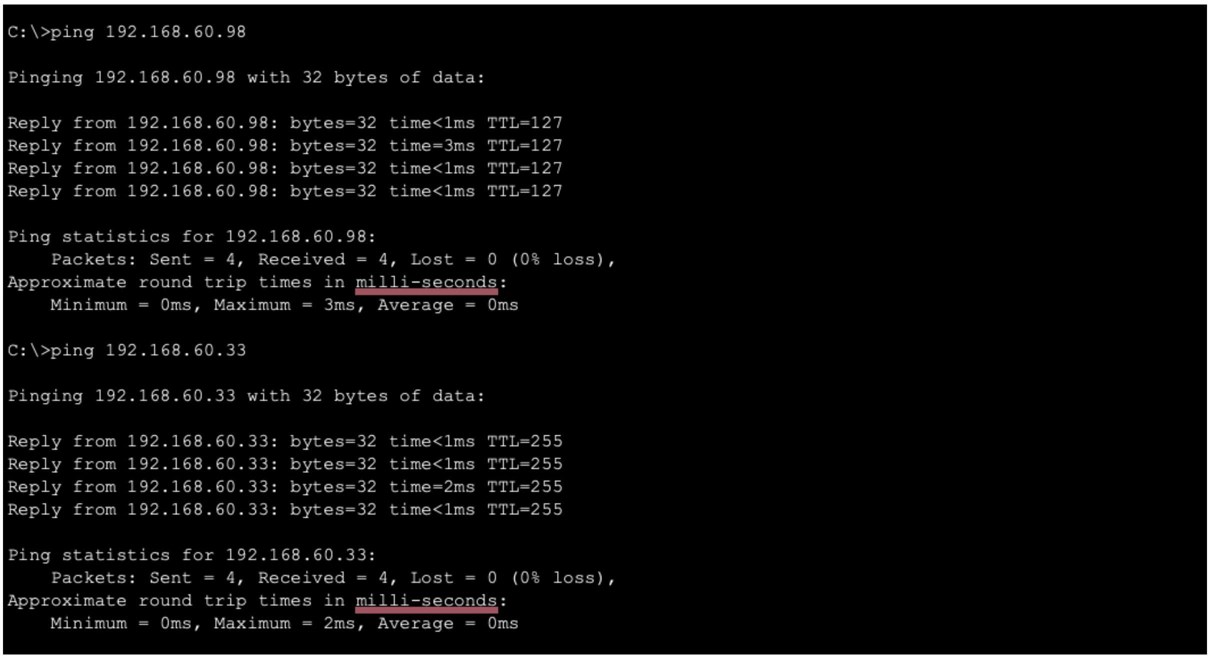
.

- Fix: switchport trunk allowed VLAN10,20,30,40,50,60,99 on core switch(SW1) Fa0/2.



```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface fa0/2
SW1(config-if)# switchport trunk allowed vlan 10,20,30,40,50,60,99
SW1(config-if)# exit
SW1(config)#end
SW1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30,40,50,60,99
Fa0/2       10,20,30,40,50,60,99
Fa0/3       10,20,30,40,50,60,99
```

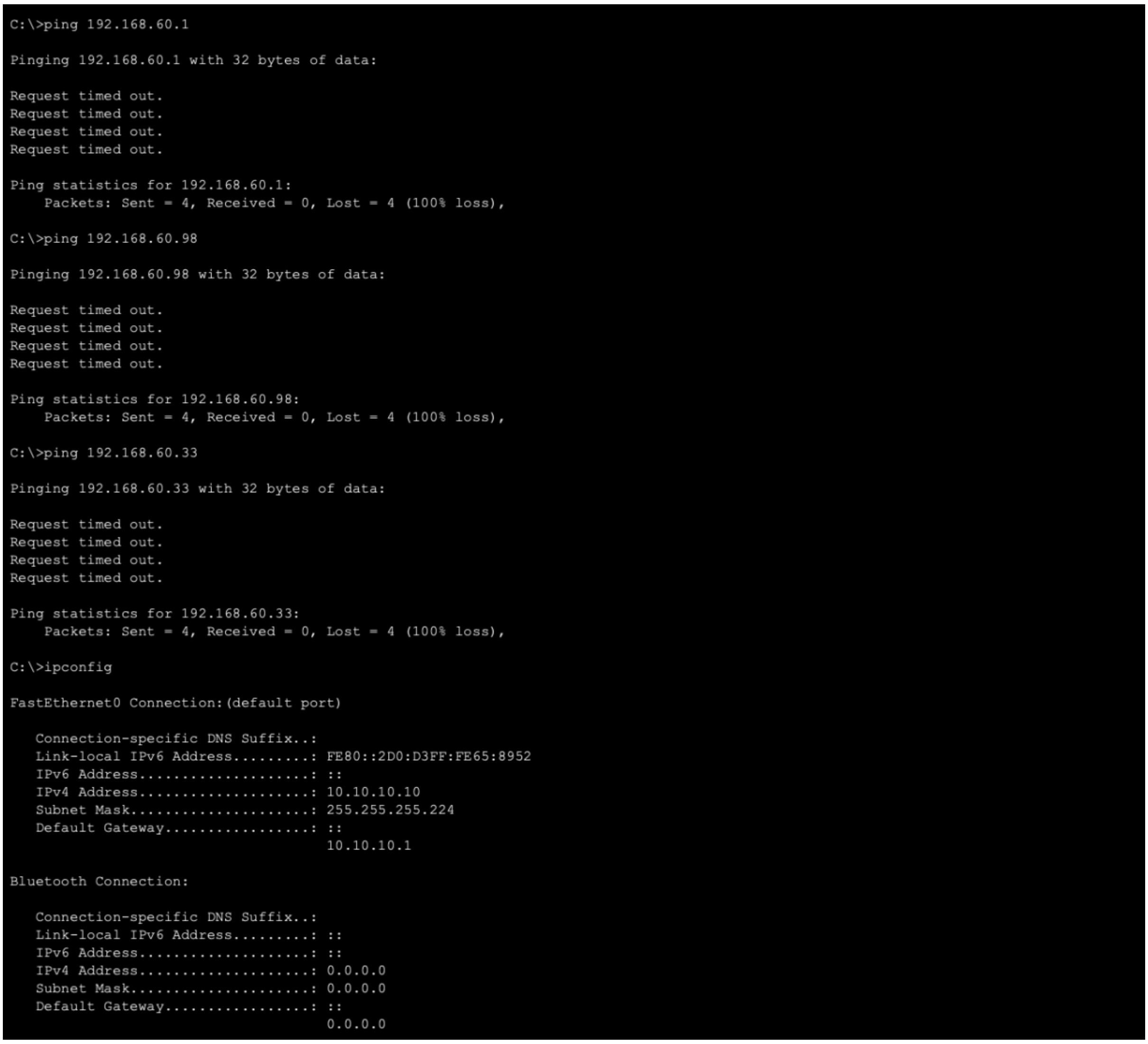
Verification: Pings to 192.168.60.33 and 192.168.60.98 are successful on PC-AN1.



```
C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time=3ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.60.33

Pinging 192.168.60.33 with 32 bytes of data:

Reply from 192.168.60.33: bytes=32 time<1ms TTL=255
Reply from 192.168.60.33: bytes=32 time<1ms TTL=255
Reply from 192.168.60.33: bytes=32 time=2ms TTL=255
Reply from 192.168.60.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

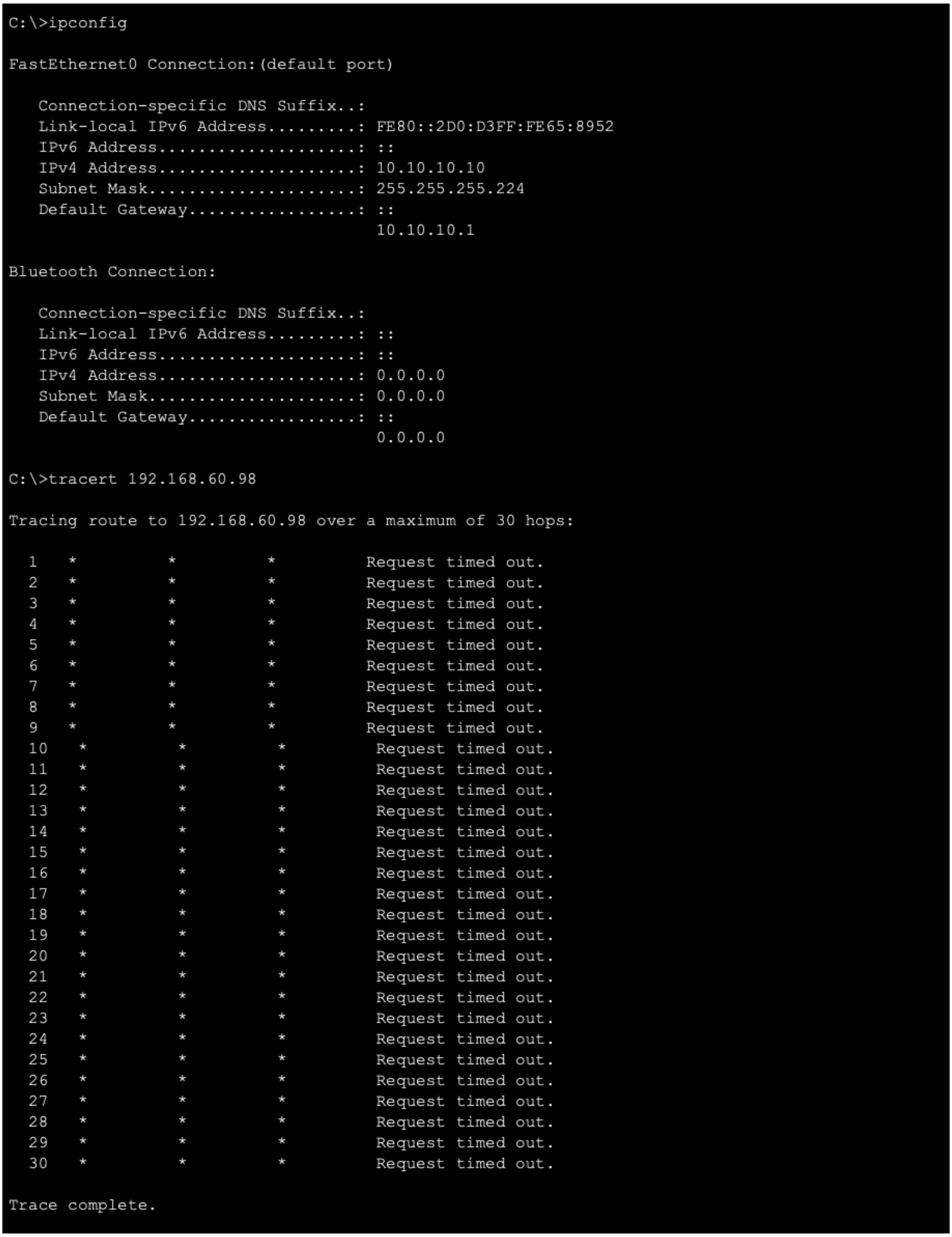
3. Fault 3:Wrong IP address on a PC.

- Symptom: PC can’t reach any device that is outside it’s subnet = PC-AN3 cannot reach its gateway (192.168.60.1), the server(192.168.60.98), and any other VLAN.



```
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.60.33

Pinging 192.168.60.33 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.60.33:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:D3FF:FE65:8952
   IPv6 Address....................: ::
   IPv4 Address....................: 10.10.10.10
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     10.10.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

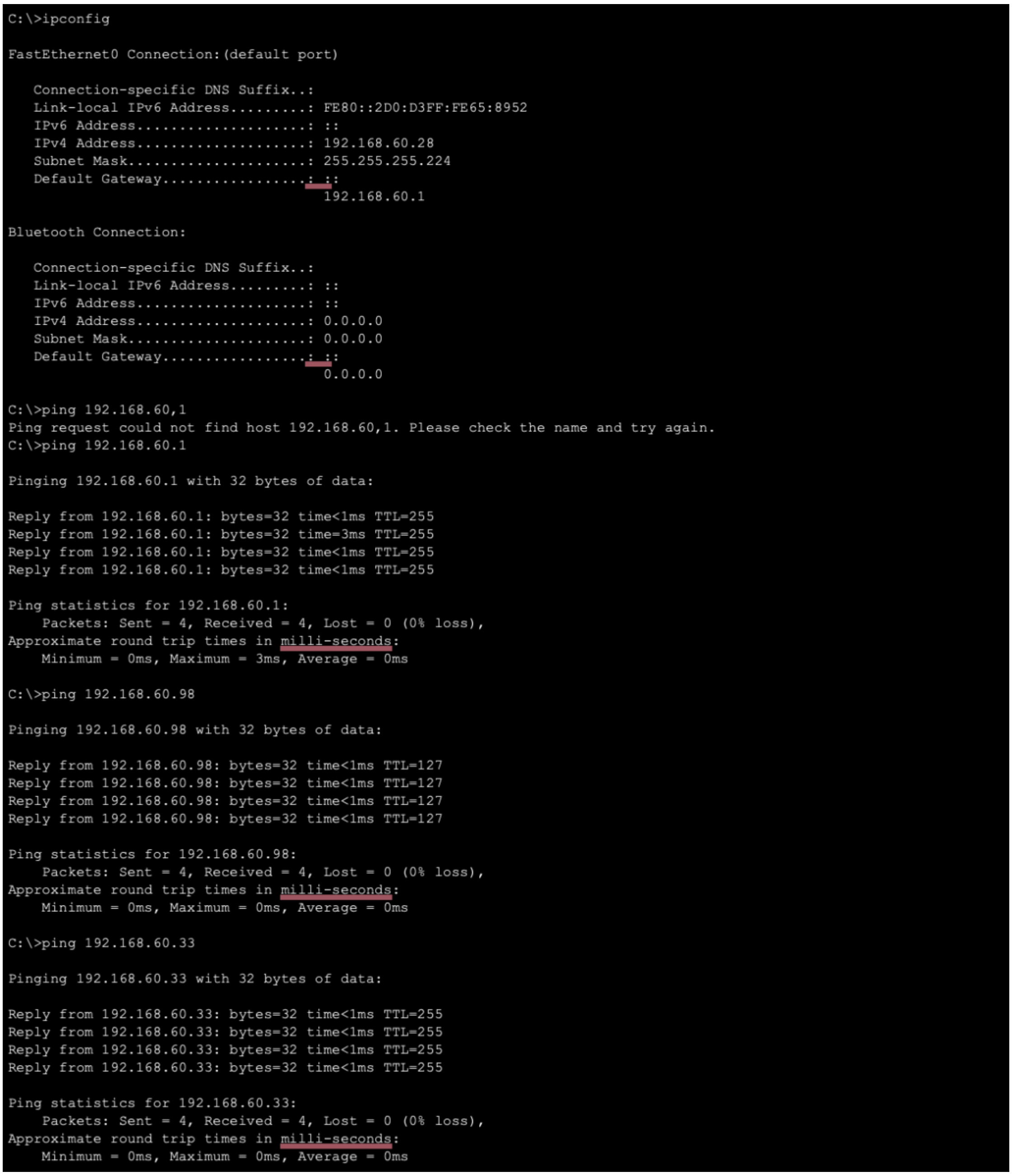
- Diagnostic: “ipconfig” on PC-AN3 shows IP 10.10.10.10 and mask 255.255.255.0. “tracert 192.168.60.98” failed at all hops.



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: 
   Link-local IPv6 Address.........: FE80::2D0:D3FF:FE65:8952
   IPv6 Address....................: ::
   IPv4 Address....................: 10.10.10.10
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     10.10.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: 
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>tracert 192.168.60.98

Tracing route to 192.168.60.98 over a maximum of 30 hops:

  1   *         *         *         Request timed out.
  2   *         *         *         Request timed out.
  3   *         *         *         Request timed out.
  4   *         *         *         Request timed out.
  5   *         *         *         Request timed out.
  6   *         *         *         Request timed out.
  7   *         *         *         Request timed out.
  8   *         *         *         Request timed out.
  9   *         *         *         Request timed out.
  10   *         *         *         Request timed out.
  11   *         *         *         Request timed out.
  12   *         *         *         Request timed out.
  13   *         *         *         Request timed out.
  14   *         *         *         Request timed out.
  15   *         *         *         Request timed out.
  16   *         *         *         Request timed out.
  17   *         *         *         Request timed out.
  18   *         *         *         Request timed out.
  19   *         *         *         Request timed out.
  20   *         *         *         Request timed out.
  21   *         *         *         Request timed out.
  22   *         *         *         Request timed out.
  23   *         *         *         Request timed out.
  24   *         *         *         Request timed out.
  25   *         *         *         Request timed out.
  26   *         *         *         Request timed out.
  27   *         *         *         Request timed out.
  28   *         *         *         Request timed out.
  29   *         *         *         Request timed out.
  30   *         *         *         Request timed out.

Trace complete.
```

- Fix: Restore the correct IP settings on PC-AN3: IP: 192.168.60.28, mask: 255.255.255.224, gateway: 192.168.60.1, DNS:192.168.60.98.

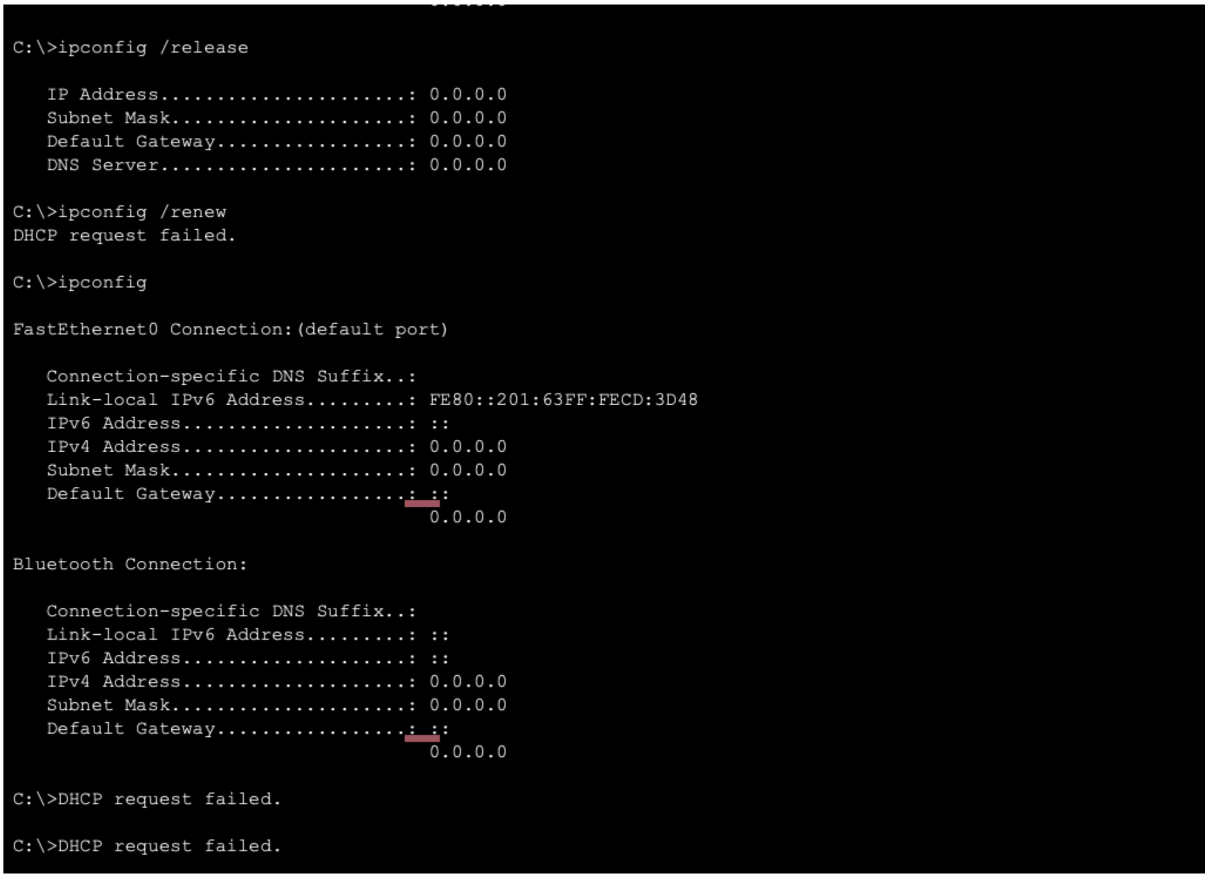
- Verification: Pings to 192.168.60.1 and 192.168.60.98 are successful on PC-AN3.



```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:D3FF:FE65:8952
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.60.28
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.60.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.60,1
Ping request could not find host 192.168.60,1. Please check the name and try again.
C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:

Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time=3ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.60.98

Pinging 192.168.60.98 with 32 bytes of data:

Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127
Reply from 192.168.60.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.60.33

Pinging 192.168.60.33 with 32 bytes of data:

Reply from 192.168.60.33: bytes=32 time<1ms TTL=255
Reply from 192.168.60.33: bytes=32 time<1ms TTL=255
Reply from 192.168.60.33: bytes=32 time<1ms TTL=255
Reply from 192.168.60.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

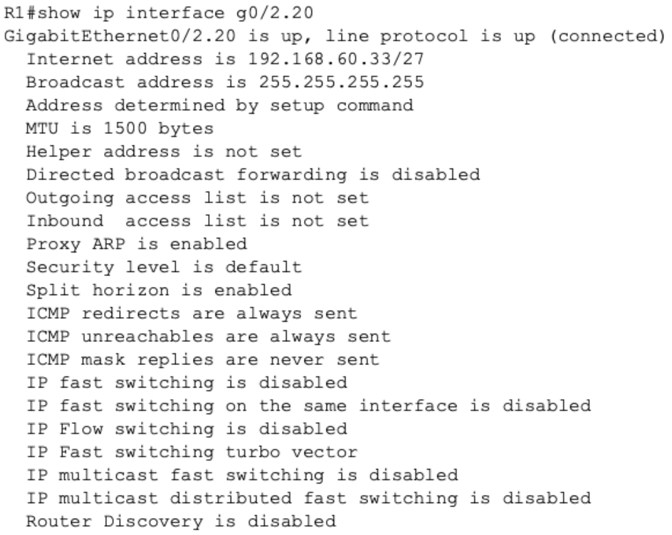
4. Fault 4: Missing IP helper address on VLAN 20 sub-interface.

- Symptom: Data Scientist PC(PC-DS3) can’t get a DHCP Address, ipconfig on PC-DS3 show 0.0.0.0 IP address.

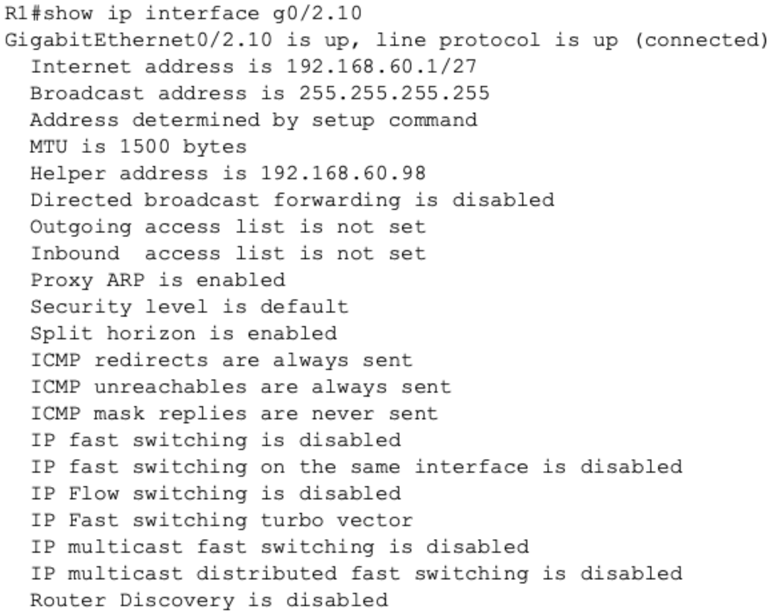


```
C:\>ipconfig /release

   IP Address......................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: 0.0.0.0
   DNS Server......................: 0.0.0.0

C:\>ipconfig /renew
DHCP request failed.

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:63FF:FECD:3D48
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>DHCP request failed.

C:\>DHCP request failed.
```

- Diagnostic: show ip interface g0/2.20 on R1 shows “ Helper address is not set” while show ip interface g0/2.10 show “ helper address is 192.168.60.98’



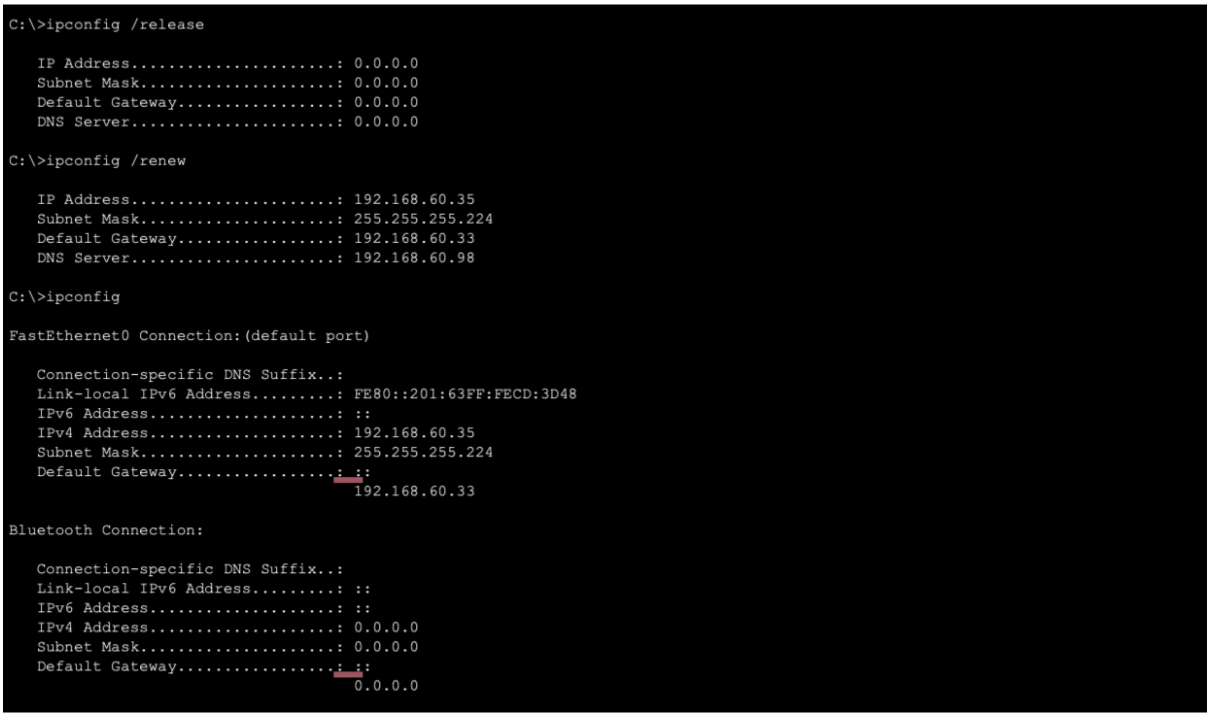
```
R1#show ip interface g0/2.20
GigabitEthernet0/2.20 is up, line protocol is up (connected)
  Internet address is 192.168.60.33/27
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
```



```
R1#show ip interface g0/2.10
GigabitEthernet0/2.10 is up, line protocol is up (connected)
  Internet address is 192.168.60.1/27
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 192.168.60.98
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
```

- Fix: Ip helper-address 192.168.60.98 on R1 interface g0/2.20

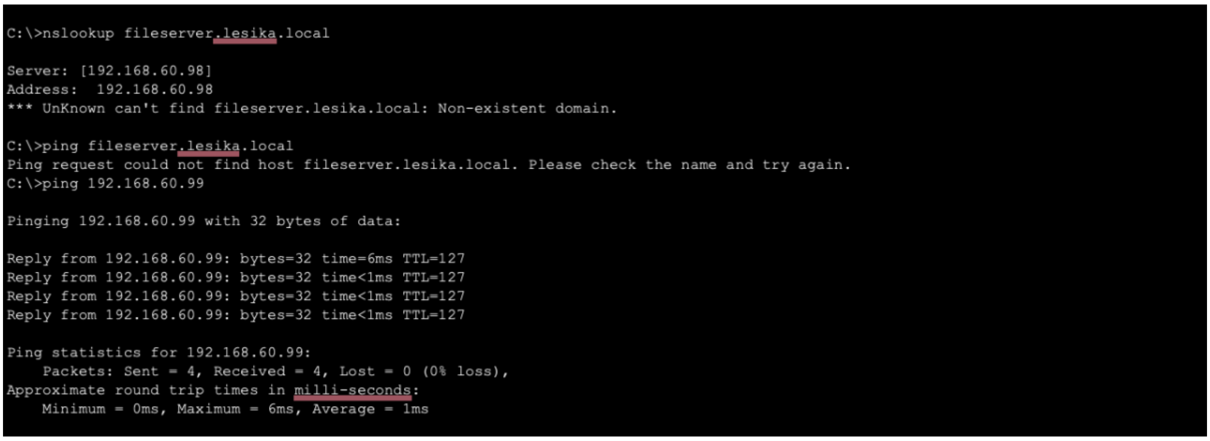
- Verification: PC-DS3 get an IP address with the range 192.168.60.34-62 with gateway 192.168.60.33



```
C:\>ipconfig /release

   IP Address......................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: 0.0.0.0
   DNS Server......................: 0.0.0.0

C:\>ipconfig /renew

   IP Address......................: 192.168.60.35
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: 192.168.60.33
   DNS Server......................: 192.168.60.98

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:63FF:FECD:3D48
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.60.35
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: ::
                                     192.168.60.33

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

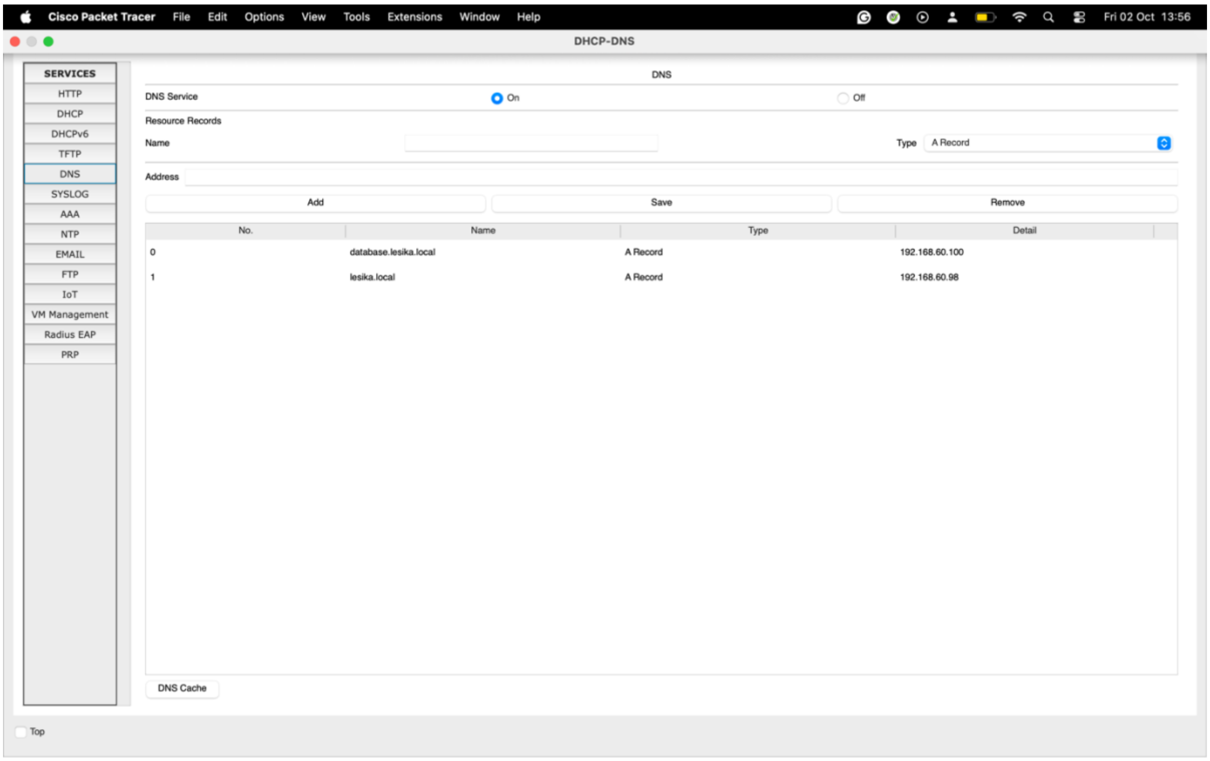
5. **Fault 5: Missing DNS A-record on the SRV-DHCP-DNS server**.

- Symptom: User(PC-AN1) can reach the server by IP 192.168.60.99, but not by hostname(fileserver.lesika.local).



```
C:\>nslookup fileserver.lesika.local

Server: [192.168.60.98]
Address:  192.168.60.98
*** UnKnown can't find fileserver.lesika.local: Non-existent domain.

C:\>ping fileserver.lesika.local
Ping request could not find host fileserver.lesika.local. Please check the name and try again.
C:\>ping 192.168.60.99

Pinging 192.168.60.99 with 32 bytes of data:

Reply from 192.168.60.99: bytes=32 time=6ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

- Diagnostic: Under Services tab of server SRV-DHCP-DNS, DNS shows no A-record of fileserver.lesika.local.


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Fri 02 Oct 13:56
DHCP-DNS
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
Add
Save
Remove
DNS Cache
Top

DNS

DNS Service ● On ○ Off

Resource Records

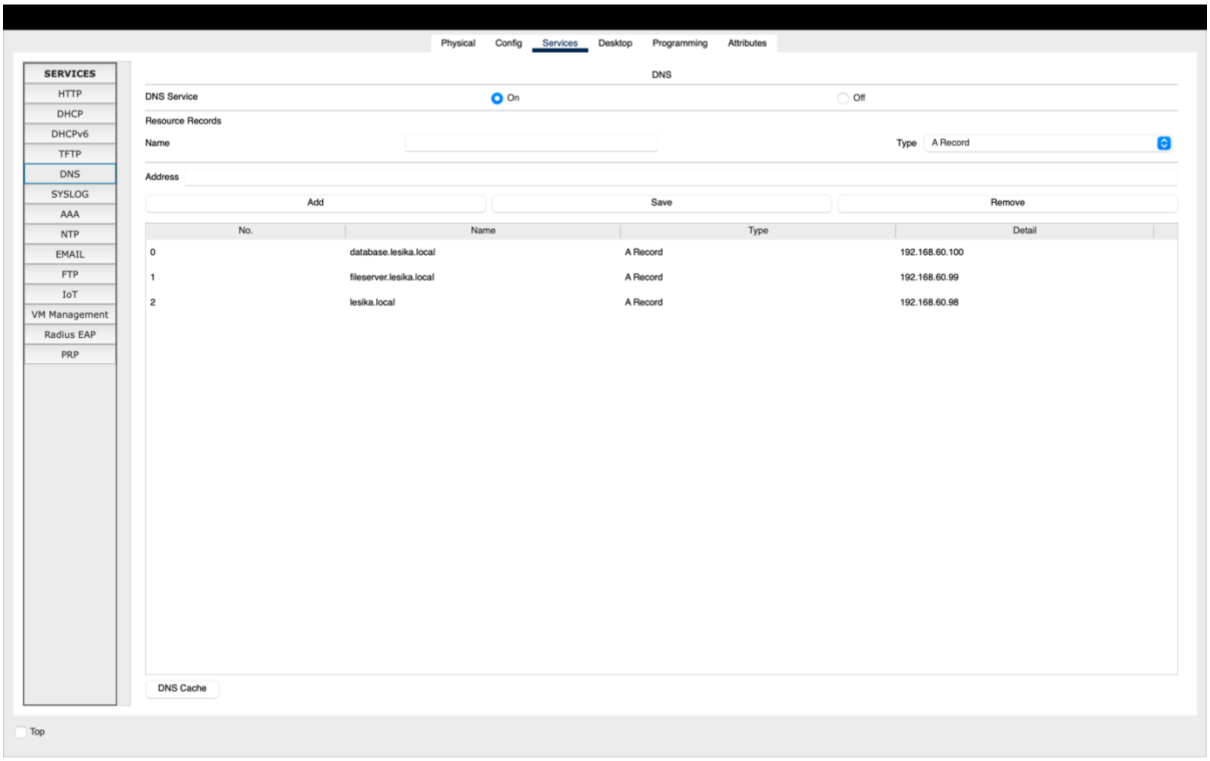
Name

Type A Record

Address

No.	Name	Type	Detail
0	database.lesika.local	A Record	192.168.60.100
1	lesika.local	A Record	192.168.60.98

- Fix: Re-adding A record of fileserver.lesika.local with Ip address 192.168.60.99.


Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
Add
Save
Remove
DNS Cache
Top

DNS

DNS Service ● On ○ Off

Resource Records

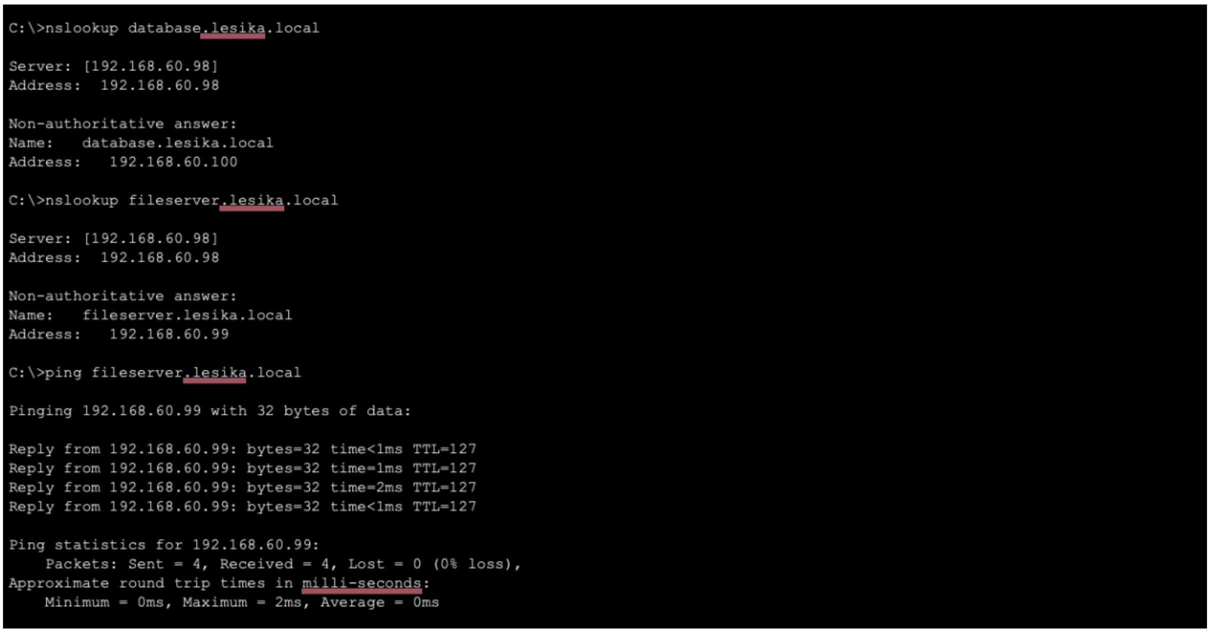
Name

Type A Record

Address

No.	Name	Type	Detail
0	database.lesika.local	A Record	192.168.60.100
1	fileserver.lesika.local	A Record	192.168.60.99
2	lesika.local	A Record	192.168.60.98

- Verification: nslookup fileserver.lesika.local fixes back to 192.168.60.99 and the nslookup fileserver.lesika.local ping succeeds.



```
C:\>nslookup database.lesika.local

Server: [192.168.60.98]
Address:  192.168.60.98

Non-authoritative answer:
Name:   database.lesika.local
Address:   192.168.60.100

C:\>nslookup fileserver.lesika.local

Server: [192.168.60.98]
Address:  192.168.60.98

Non-authoritative answer:
Name:   fileserver.lesika.local
Address:   192.168.60.99

C:\>ping fileserver.lesika.local

Pinging 192.168.60.99 with 32 bytes of data:

Reply from 192.168.60.99: bytes=32 time<1ms TTL=127
Reply from 192.168.60.99: bytes=32 time=1ms TTL=127
Reply from 192.168.60.99: bytes=32 time=2ms TTL=127
Reply from 192.168.60.99: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.60.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```